THE ENTROPY JKO MAP IS NONEXPANSIVE ON GAUSSIAN INPUTS

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Abstract. We prove that the entropy JKO map is nonexpansive on Gaussian inputs.

Notation and conventions

Symbol	Meaning
$\mathcal{P}_2(\mathbb{R}^d)$	Borel probability measures on $\mathbb{R}^d$ with finite second moment.
$\mathcal{W}_2$	The quadratic Wasserstein distance.
$\mathbb{S}^d$	Real symmetric $d \times d$ matrices.
$\mathbb{S}^d_{++}$	Real symmetric positive-definite $d \times d$ matrices.
$\mathcal{G}_d$	Nondegenerate Gaussian probability measures on $\mathbb{R}^d$.
J_τ^{Ent}	The entropy JKO resolvent on $\mathcal{P}_2(\mathbb{R}^d)$.
d_{BW}	The Bures–Wasserstein distance on $\mathbb{S}^d_{++}$.
Φ_τ	The covariance update of the entropy JKO resolvent at a Gaussian input.
$\mathcal{L}_A$	The Lyapunov operator $X \mapsto AX + XA$.
$g_A,\ \lVert\cdot\rVert_A$	The Bures–Wasserstein Riemannian metric and its induced norm at A.
$f^{[1]}$	The first divided difference of a scalar function f.

1. Introduction

Proximal mappings are a basic tool in convex optimization.

Let H be a real Hilbert space, and let $F : H \to (-\infty, +\infty]$ be proper, lower semicontinuous, and convex. For $\tau > 0$, its proximal map is defined by

$$\operatorname{prox}_{\tau F}(x) := \operatorname*{argmin}_{y \in H} \left\{ F(y) + \frac{1}{2\tau} \lVert x - y \rVert^2 \right\}.$$

The addition of the squared-distance term makes the objective strongly convex and coercive. Hence the minimization problem admits a unique solution for every $x \in H$.

Moreover, the proximal map is firmly nonexpansive:

$$\lVert \operatorname{prox}_{\tau F}(x) - \operatorname{prox}_{\tau F}(y) \rVert^2 \le \langle \operatorname{prox}_{\tau F}(x) - \operatorname{prox}_{\tau F}(y), x - y \rangle .$$

In particular, $\lVert \operatorname{prox}_{\tau F}(x) - \operatorname{prox}_{\tau F}(y) \rVert \le \lVert x - y \rVert$. Thus, the proximal map is nonexpansive. Firm nonexpansivity follows from the monotonicity of the subdifferential ∂F; see [BC17, Proposition 12.28].

The Jordan–Kinderlehrer–Otto scheme [JKO98] carries the same variational idea to the Wasserstein space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$. Given a functional

$$\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \to (-\infty, +\infty],$$

one considers its Wasserstein proximal set

$$J_\tau^{\mathcal{F}}(\mu) := \operatorname*{argmin}_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \mathcal{F}(\nu) + \frac{1}{2\tau} \mathcal{W}_2^2(\nu, \mu) \right\}. \tag{1.1}$$

Here the right-hand side is a priori a set of minimizers. Its existence and uniqueness require additional assumptions on $\mathcal{F}$.

Two related questions naturally arise. Under what assumptions on $\mathcal{F}$ is $J_\tau^{\mathcal{F}}(\mu)$ nonempty and a singleton for every $\mu \in \mathcal{P}_2(\mathbb{R}^d)$? Once the minimizer is unique, when does the resulting resolvent satisfy

$$\mathcal{W}_2(J_\tau^{\mathcal{F}}(\mu_0), J_\tau^{\mathcal{F}}(\mu_1)) \le \mathcal{W}_2(\mu_0, \mu_1)?$$

In dimension $d = 1$, the quantile representation identifies $\mathcal{P}_2(\mathbb{R})$ isometrically with a convex subset of $L^2(0,1)$. Consequently, the proximal map of a geodesically convex functional is nonexpansive. But this Hilbertian argument is no longer available in dimensions $d > 1$.

Recent work of Di Marino, Farinelli, and Naldi [DMFN26] studies the stability of Wasserstein proximal maps under several notions of convexity. They show that total convexity implies nonexpansivity of the proximal map. They also introduce convexity along 2-base generalized geodesics and prove that, if $\mathcal{F}$ is λ-convex in this sense, then $(1+\lambda\tau)\mathcal{W}_2(J_\tau^{\mathcal{F}}(\mu_0), J_\tau^{\mathcal{F}}(\mu_1)) \le \mathcal{W}_2(\mu_0,\mu_1)$, provided that $1+\lambda\tau > 0$. For ordinary geodesic convexity, however, they obtain only weaker forms of stability. Thus, their results give useful sufficient conditions for nonexpansivity, but they do not settle whether geodesic convexity alone is sufficient in dimensions $d > 1$.

A canonical geodesically convex functional on Wasserstein space is the Boltzmann entropy; see, for example, [AGS08, Remark 9.3.10]. It is defined by

$$\operatorname{Ent}(\mu) := \begin{cases} \displaystyle\int_{\mathbb{R}^d} \rho(x)\log\rho(x)\,dx, & \text{if } d\mu = \rho\,dx, \\ +\infty, & \text{otherwise.} \end{cases}$$

Despite its geodesic convexity, the entropy lies outside both stronger convexity classes discussed above. Indeed, it is not totally convex; see [DMFN26, Corollary 2.14]. Nor is it convex along 2-base generalized geodesics; see [DMFN26, Example 3.18]. Therefore, the entropy does not satisfy either of their sufficient conditions for ordinary $\mathcal{W}_2$-nonexpansivity. Consequently, the criteria established in [DMFN26] do not determine whether the entropy resolvent is nonexpansive. To the best of our knowledge, ordinary $\mathcal{W}_2$-nonexpansivity of the entropy JKO map remains open in the Wasserstein space.

In this paper, we do not attempt to solve this question for arbitrary pairs of inputs in the Wasserstein space. Instead, we restrict the inputs to the family of nondegenerate Gaussian measures

$$\mathcal{G}_d := \left\{\mathcal{N}(m,A) : m \in \mathbb{R}^d,\ A \in \mathbb{S}^d_{++}\right\}.$$

Through the parametrization $(m,A) \mapsto \mathcal{N}(m,A)$, the family $\mathcal{G}_d$ is naturally identified with the finite-dimensional manifold $\mathbb{R}^d \times \mathbb{S}^d_{++}$.

Importantly, only the inputs are restricted. For $\mu \in \mathcal{G}_d$, we keep the entropy JKO resolvent

$$J_\tau^{\operatorname{Ent}}(\mu) := \operatorname*{argmin}_{\nu\in\mathcal{P}_2(\mathbb{R}^d)} \left\{\operatorname{Ent}(\nu) + \frac{1}{2\tau}\mathcal{W}_2^2(\nu,\mu)\right\}.$$

We show in Section 2 that replacing any competitor by the Gaussian measure with the same mean and covariance cannot increase this objective. Thus, at a Gaussian input, the minimization is equivalent to minimization over $\mathcal{G}_d$. The latter problem is known to have a unique minimizer whose covariance is given by an explicit closed-form formula; see [DBCS23, Lemma 3.2] and also [Wib18, Example 7]. The explicit formula describes each individual JKO step, but it does not by itself provide a comparison between the outputs corresponding to two different Gaussian inputs. We therefore ask whether the entropy JKO resolvent is nonexpansive between Gaussian inputs. Our main result gives an affirmative answer.

Theorem 1.1 (Main result). *For every $\tau > 0$, the entropy JKO resolvent satisfies*

$$\mathcal{W}_2(J_\tau^{\operatorname{Ent}}(\mu_0), J_\tau^{\operatorname{Ent}}(\mu_1)) \le \mathcal{W}_2(\mu_0,\mu_1)$$

for all $\mu_0,\mu_1 \in \mathcal{G}_d$. More precisely, if $\mu_i = \mathcal{N}(m_i,A_i)$, then the inequality is strict when $A_0 \ne A_1$, while equality holds when $A_0 = A_1$.

2. Gaussianization and reduction to a covariance map

We first show that, at a Gaussian input, the entropy JKO problem can be reduced without loss to Gaussian competitors. We then use two standard formulas for Gaussian measures and invoke the explicit solution of the resulting covariance minimization problem from [DBCS23, Lemma 3.2]; see also [Wib18, Example 7].

2.1. **Gaussianization of the JKO problem.**

For $\nu \in \mathcal{P}_2(\mathbb{R}^d)$, let $\mathsf{G}\nu$ denote the Gaussian measure with the same mean and covariance as ν, whenever the covariance is positive definite.

Proposition 2.1 (Gaussianization). *Let $\tau > 0$ and $\mu \in \mathcal{G}_d$. Then*

$$\operatorname*{argmin}_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \operatorname{Ent}(\nu) + \frac{1}{2\tau} \mathcal{W}_2^2(\nu, \mu) \right\} = \operatorname*{argmin}_{\nu \in \mathcal{G}_d} \left\{ \operatorname{Ent}(\nu) + \frac{1}{2\tau} \mathcal{W}_2^2(\nu, \mu) \right\}.$$

In particular, the entropy JKO minimizer at a nondegenerate Gaussian input is itself a nondegenerate Gaussian measure.

Proof. Consider first a competitor $\nu \in \mathcal{P}_2(\mathbb{R}^d)$ with finite entropy. Then by definition, ν is absolutely continuous, and its covariance is positive definite: otherwise one can prove that ν would be supported on a proper affine hyperplane. For this affine hyperplane $\mathcal{H}$, $\nu(H) = 1$ but the Lebesgue measure of $\mathcal{H}$ is 0. Thus $\mathsf{G}\nu \in \mathcal{G}_d$ is well defined.

Because ν and $\mathsf{G}\nu$ have the same first two moments, the maximum-entropy property of Gaussian measures ([CT06, Theorem 8.6.5]) gives the exact identity

$$\operatorname{Ent}(\nu) - \operatorname{Ent}(\mathsf{G}\nu) =\geq 0,$$

Moreover, the Gelbrich inequality [Gel90] and the identity $\mathsf{G}\mu = \mu$ give

$$\mathcal{W}_2(\mathsf{G}\nu, \mu) = \mathcal{W}_2(\mathsf{G}\nu, \mathsf{G}\mu) \leq \mathcal{W}_2(\nu, \mu).$$

Hence replacing ν by $\mathsf{G}\nu$ cannot increase the JKO objective. Competitors with infinite entropy cannot improve the infimum, so the two minimization problems have the same value.

The Gaussian problem has a unique minimizer by [DBCS23, Lemma 3.2]. It is also a minimizer of the problem. Conversely, if ν belongs to LHS. Since two minimization problems have the same value, it requires $\operatorname{Ent}(\nu) - \operatorname{Ent}(\mathsf{G}\nu) = 0$, hence $\nu = \mathsf{G}\nu$, again by the the maximum-entropy property of Gaussian measures. Therefore every minimizer is Gaussian, and the two argmin sets coincide. □

It remains to solve the Gaussian minimization problem. Let $\mu = \mathcal{N}(m, A) \in \mathcal{G}_d, \nu = \mathcal{N}(n, B) \in \mathcal{G}_d$. We proceed in three steps.

2.2. **Entropy and Wasserstein distance of Gaussian measures.**

First, the entropy of ν is

$$\operatorname{Ent}\big(\mathcal{N}(n, B)\big) = -\frac{1}{2} \log \det B + C_d, \tag{2.1}$$

where C_d depends only on the dimension. In particular, the entropy is independent of the mean; see, for example, [Wib18, Appendix E.2].

Second, the quadratic Wasserstein distance between two nondegenerate Gaussian measures satisfies

$$\mathcal{W}_2^2(\mathcal{N}(n, B), \mathcal{N}(m, A)) = \|n - m\|^2 + d_{\mathrm{BW}}^2(B, A), \tag{2.2}$$

where

$$d_{\mathrm{BW}}^2(B, A) := \operatorname{tr}(B) + \operatorname{tr}(A) - 2 \operatorname{tr}\left(B^{1/2} A B^{1/2}\right)^{1/2}. \tag{2.3}$$

This is the Bures–Wasserstein distance on $\mathbb{S}_{++}^d$; see, for example, [Gel90, Tak11].

2.3. **Separation of the mean and covariance.** Substituting (2.1) and (2.2) into the definition of J_τ^{Ent}, using Proposition 2.1, gives

$$\begin{aligned}\mathrm{Ent}(\mathcal{N}(n,B)) &+ \frac{1}{2\tau}\mathcal{W}_2^2(\mathcal{N}(n,B),\mathcal{N}(m,A)) \\ &= C_d + \frac{1}{2\tau}\|n-m\|^2 + \left[\frac{1}{2\tau}d_{\mathrm{BW}}^2(B,A) - \frac{1}{2}\log\det B\right]. \end{aligned} \tag{2.4}$$

The variable n appears only in the nonnegative term $\|n-m\|^2/(2\tau)$. Its unique minimizer is therefore $n=m$. Consequently, we see the JKO step preserves the mean, and the original minimization problem reduces to

$$\Phi_\tau(A) := \operatorname*{argmin}_{B\in\mathbb{S}_{++}^d}\left\{\frac{1}{2\tau}d_{\mathrm{BW}}^2(B,A) - \frac{1}{2}\log\det B\right\}. \tag{2.5}$$

2.4. **The explicit covariance update.** The minimization problem (2.5) has a unique solution. By [DBCS23, Lemma 3.2] and [Wib18, Example 7], it is given by

$$\Phi_\tau(A) = \frac{1}{2}\left(A + 2\tau I + (A(A+4\tau I))^{1/2}\right). \tag{2.6}$$

Equivalently, if $f_\tau(s) := \frac{1}{2}\left(s + 2\tau + \sqrt{s(s+4\tau)}\right)$ $(s>0)$, then

$$\Phi_\tau(A) = f_\tau(A) \tag{2.7}$$

in the sense of spectral functional calculus.

We have thus obtained the following simpler representation of the map defined in Section 1.

Proposition 2.2 (Explicit Gaussian JKO step)**.** *For every $\tau>0$ and every $\mathcal{N}(m,A)\in\mathcal{G}_d$,*

$$J_\tau^{\mathrm{Ent}}(\mathcal{N}(m,A)) = \mathcal{N}(m,\Phi_\tau(A)),$$

where Φ_τ is the spectral map given by (2.6).

In particular, for $\mu_i = \mathcal{N}(m_i,A_i)\in\mathcal{G}_d$, $i=0,1$,

$$\begin{aligned}\mathcal{W}_2^2(&J_\tau^{\mathrm{Ent}}(\mu_0), J_\tau^{\mathrm{Ent}}(\mu_1)) \\ &= \|m_0-m_1\|^2 + d_{\mathrm{BW}}^2(\Phi_\tau(A_0),\Phi_\tau(A_1)).\end{aligned}$$

Therefore, to prove Theorem 1.1, it suffices to establish the matrix inequality

$$d_{\mathrm{BW}}(\Phi_\tau(A_0),\Phi_\tau(A_1)) \le d_{\mathrm{BW}}(A_0,A_1).$$

For the strict part of the theorem, we will also show that this inequality is strict when $A_0\neq A_1$.

3. Proof of the covariance inequality

In this section we prove the matrix inequality obtained at the end of Section 2. The proof has two parts. First, we give a general criterion for a spectral matrix map to be nonexpansive in the Bures–Wasserstein distance. Second, we verify this criterion for the scalar function f_τ in (2.7).

3.1. **A differential criterion for spectral maps.** We first recall two standard formulas.

Let $\mathbb{S}^d$ be the space of real symmetric $d\times d$ matrices. Since $\mathbb{S}_{++}^d$ is an open submanifold of $\mathbb{S}^d$, its tangent space at every A is naturally identified with $\mathbb{S}^d$. Set $\mathcal{L}_A(X) := AX + XA$. On the manifold $\mathbb{S}_{++}^d$, the Bures–Wasserstein distance is the length distance of the Riemannian metric

$$\begin{aligned} g_A\colon T_A\mathbb{S}_{++}^d \times T_A\mathbb{S}_{++}^d &\longrightarrow \mathbb{R} \\ (H,K) &\longmapsto \frac{1}{2}\operatorname{tr}(\mathcal{L}_A^{-1}(H)K). \end{aligned} \tag{3.1}$$

If $A = Q\,\mathrm{diag}(\lambda_1, \dots, \lambda_d)Q^\mathsf{T}$ and $\widehat{H} = Q^\mathsf{T} H Q$, then the induced norm is

$$\|H\|_A^2 := g_A(H, H) = \frac{1}{2}\sum_{i,j=1}^{d} \frac{\widehat{H}_{ij}^2}{\lambda_i + \lambda_j}. \tag{3.2}$$

Both (3.1) and (3.2) are standard; see [Tak11].

For a C^1 function $f : (0, \infty) \to (0, \infty)$, write

$$f^{[1]}(x, y) := \begin{cases} \dfrac{f(x) - f(y)}{x - y}, & x \neq y, \\ f'(x), & x = y. \end{cases}$$

If $F(A) = f(A)$ is the associated spectral matrix map, the Daleckii–Krein formula gives

$$Q^\mathsf{T} DF(A)[H]Q = \left(f^{[1]}(\lambda_i, \lambda_j)\widehat{H}_{ij}\right)_{i,j=1}^{d}; \tag{3.3}$$

see [Hig08, Chapter 3]. This formula also includes the noncommuting, off-diagonal directions $i \neq j$.

Proposition 3.1 (Differential nonexpansivity criterion)**.** *Suppose that*

$$\left(f^{[1]}(x, y)\right)^2 (x + y) \leq f(x) + f(y) \qquad \text{for all } x, y > 0. \tag{3.4}$$

Then the spectral map $F(A) = f(A)$ satisfies

$$\|DF(A)[H]\|_{F(A)} \leq \|H\|_A \tag{3.5}$$

for every $A \in \mathbb{S}_{++}^d$ and $H \in \mathbb{S}^d$.

Consequently,

$$d_{\mathrm{BW}}(F(A), F(B)) \leq d_{\mathrm{BW}}(A, B) \tag{3.6}$$

for all $A, B \in \mathbb{S}_{++}^d$. If the inequality in (3.4) is strict for all $x, y > 0$, then

$$d_{\mathrm{BW}}(F(A), F(B)) < d_{\mathrm{BW}}(A, B) \qquad \text{whenever } A \neq B.$$

Proof. Diagonalize A as above and denote $\widehat{H} = Q^\mathsf{T} H Q$. Since the eigenvalues of $F(A)$ are $f(\lambda_i)$, formulas (3.2) and (3.3) give

$$\begin{aligned} \|DF(A)[H]\|_{F(A)}^2 &= \frac{1}{2}\sum_{i,j=1}^{d} \frac{\left(f^{[1]}(\lambda_i, \lambda_j)\right)^2 \widehat{H}_{ij}^2}{f(\lambda_i) + f(\lambda_j)} \\ &\leq \frac{1}{2}\sum_{i,j=1}^{d} \frac{\widehat{H}_{ij}^2}{\lambda_i + \lambda_j} \\ &= \|H\|_A^2. \end{aligned} \tag{3.7}$$

Here the inequality is exactly (3.4), applied with $(x, y) = (\lambda_i, \lambda_j)$. This proves the differential estimate.

Now let $\gamma : [0, 1] \to \mathbb{S}_{++}^d$ be any piecewise C^1 curve from A to B. Thus, the following integrals are taken along the Riemannian manifold $\mathbb{S}_{++}^d$, with $\dot{\gamma}(t) \in T_{\gamma(t)}\mathbb{S}_{++}^d$. The chain rule gives

$$\begin{aligned} L_{\mathrm{BW}}(F \circ \gamma) &= \int_0^1 \|DF(\gamma(t))[\dot{\gamma}(t)]\|_{F(\gamma(t))}\, dt \\ &\leq \int_0^1 \|\dot{\gamma}(t)\|_{\gamma(t)}\, dt = L_{\mathrm{BW}}(\gamma). \end{aligned} \tag{3.8}$$

Taking the infimum over all such curves proves (3.6).

Suppose now that the scalar inequality in (3.4) is strict. Let us define the continuous function $\Phi : (0,\infty)\times(0,\infty)\to\mathbb{R}$ by

$$\Phi(x,y) = \frac{\left(f^{[1]}(x,y)\right)^2 (x+y)}{f(x)+f(y)}.$$

By our strict hypothesis, $\Phi(x,y) < 1$ for all $x, y > 0$.

Let $\gamma : [0,1] \to \mathbb{S}^d_{++}$ be a minimizing Bures–Wasserstein geodesic from A to B (which exists entirely within $\mathbb{S}^d_{++}$; see [Tak11]). By the continuity of the geodesic γ, its trajectory $\gamma([0,1])$ is a compact subset of $\mathbb{S}^d_{++}$. Consider the mapping that assigns to each symmetric matrix its spectrum. By classical matrix analysis, this mapping is continuous. Therefore, the image of the compact set $\gamma([0,1])$ under this spectral mapping, given by

$$K := \bigcup_{t\in[0,1]} \operatorname{Spec}(\gamma(t)),$$

is a compact subset of $\mathbb{R}$. Furthermore, because $\gamma([0,1])$ lies entirely within $\mathbb{S}^d_{++}$, every element of K is strictly positive. Consequently, K is a compact subset of $(0,\infty)$.

Since $K\times K$ is compact and Φ is continuous, Φ does attain its maximum on $K \times K$. We define

$$q_*^2 = \max_{(x,y)\in K\times K} \Phi(x,y).$$

Because the strict inequality holds globally, we strictly have $q_*^2 < 1$, hence $q_* < 1$.

Now, for any $C = \gamma(t)$ on the geodesic and any $H \in \mathbb{S}^d$, diagonalizing C as before shows that the coefficients in the local metric expansion are exactly $\Phi(\lambda_i, \lambda_j)$ with $\lambda_i, \lambda_j \in K$. Thus, bounded by q_*^2, we obtain the uniform bound

$$\|DF(C)[H]\|_{F(C)} \le q_* \|H\|_C \,.$$

If $A \neq B$, integrating along the geodesic gives

$$d_{\mathrm{BW}}(F(A),F(B)) \le L_{\mathrm{BW}}(F\circ\gamma) \le q_* L_{\mathrm{BW}}(\gamma) < d_{\mathrm{BW}}(A,B),$$

which completes the proof. □

3.2. **The scalar inequality for the entropy update.** We now apply Proposition 3.1 to

$$f_\tau(x) = \frac{1}{2}\left(x + 2\tau + \sqrt{x(x+4\tau)}\right). \tag{3.9}$$

The next lemma contains the main calculation of the proof.

Lemma 3.2 (Key scalar inequality)**.** *For every $\tau > 0$ and $x, y > 0$,*

$$\left(f_\tau^{[1]}(x,y)\right)^2 (x+y) < f_\tau(x) + f_\tau(y). \tag{3.10}$$

Proof. Set $u = f_\tau(x)$, $v = f_\tau(y)$, $a = u/\tau > 1$, and $b = v/\tau > 1$.

From the scalar formula for f_τ,

$$2u - x - 2\tau = \sqrt{x(x+4\tau)}.$$

Squaring both sides gives

$$4u^2 - 4u(x+2\tau) + (x+2\tau)^2 = x^2 + 4\tau x.$$

Since $(x+2\tau)^2 = x^2 + 4\tau x + 4\tau^2$, cancellation yields

$$u^2 - u(x+2\tau) + \tau^2 = 0.$$

Solving this identity for x gives $x = u - 2\tau + \tau^2/u$. The same calculation with $v = f_\tau(y)$ gives $y = v - 2\tau + \tau^2/v$. Thus

$$x = \tau\left(a + \frac{1}{a} - 2\right), \qquad y = \tau\left(b + \frac{1}{b} - 2\right), \tag{3.11}$$

If $x \neq y$, subtracting the two identities in (3.11) gives

$$x - y = (u - v)\left(1 - \frac{\tau^2}{uv}\right).$$

Therefore,

$$f_\tau^{[1]}(x, y) = \frac{u - v}{x - y} = \frac{uv}{uv - \tau^2} = \frac{ab}{ab - 1}.$$

If $x = y$, differentiating $x = u - 2\tau + \tau^2/u$ gives $f'_\tau(x) = (1 - \tau^2/u^2)^{-1} = a^2/(a^2 - 1)$, which is the same formula with $a = b$. Consequently,

$$x + y = \tau\left(a + b + \frac{1}{a} + \frac{1}{b} - 4\right), \qquad f_\tau^{[1]}(x, y) = \frac{ab}{ab - 1}. \tag{3.12}$$

Now write $a = 1 + r$ and $b = 1 + s$, where $r, s > 0$. A direct calculation, first with a common denominator and then with this substitution, gives

$$\begin{aligned}
&\frac{1}{\tau}\left[f_\tau(x) + f_\tau(y) - \left(f_\tau^{[1]}(x, y)\right)^2 (x + y)\right] \\
&= a + b - \frac{a^2 b^2}{(ab - 1)^2}\left(a + b + \frac{1}{a} + \frac{1}{b} - 4\right) \\
&= \frac{(a + b)(ab - 1)^2 - a^2 b^2 \left(a + b + \frac{1}{a} + \frac{1}{b} - 4\right)}{(ab - 1)^2} \\
&= \frac{4a^2 b^2 - 3a^2 b - 3ab^2 + a + b}{(ab - 1)^2} \\
&= \frac{r^2 + s^2 + 4rs + 5r^2 s + 5rs^2 + 4r^2 s^2}{(ab - 1)^2} > 0.
\end{aligned}$$

The last numerator is positive because $r, s > 0$. This is exactly (3.10). $\square$

3.3. **Proof of the main theorem.**

Proof of Theorem 1.1. By Lemma 3.2, the scalar function f_τ satisfies the hypothesis of Proposition 3.1. Since $\Phi_\tau(A) = f_\tau(A)$, (3.6) gives

$$d_{\mathrm{BW}}(\Phi_\tau(A_0), \Phi_\tau(A_1)) \leq d_{\mathrm{BW}}(A_0, A_1) \tag{3.13}$$

for all $A_0, A_1 \in \mathbb{S}^d_{++}$. Moreover, the strict part of Proposition 3.1 shows that the inequality is strict when $A_0 \neq A_1$.

It remains only to return to Gaussian measures. Let $\mu_i = \mathcal{N}(m_i, A_i)$, $i = 0, 1$. By Proposition 2.2, the JKO step preserves each mean and replaces A_i by $\Phi_\tau(A_i)$. Hence

$$\begin{aligned}
&\mathcal{W}_2^2(J_\tau^{\mathrm{Ent}}(\mu_0), J_\tau^{\mathrm{Ent}}(\mu_1)) \\
&\quad = \|m_0 - m_1\|^2 + d_{\mathrm{BW}}^2(\Phi_\tau(A_0), \Phi_\tau(A_1)) \\
&\quad \leq \|m_0 - m_1\|^2 + d_{\mathrm{BW}}^2(A_0, A_1) \\
&\quad = \mathcal{W}_2^2(\mu_0, \mu_1).
\end{aligned} \tag{3.14}$$

If $A_0 \neq A_1$, the middle inequality is strict. If $A_0 = A_1$, both covariance terms vanish, so equality holds. This proves all parts of the theorem. $\square$

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